Glossary: Can't Ship AI Without Knowing If It Helps (or Hurts!)

Data Quality & its Role

- **Accuracy:** Values correct; wrong inputs → wrong predictions.
- **Completeness:** No missing values; gaps cause model failure.
- **Timeliness:** Data current; stale data degrades real-time AI.
- **Consistency:** Same meaning across sources.
- **Validity:** Conforms to rules/formats; invalid data fails validation (Pydantic, GX).

Metrics: Processing & Operational

- **Throughput:** Records/requests processed per unit time.
- **Latency (p50/p95):** Response time at 50th/95th percentile.
- **Ingestion rate:** Rate at which data enters the pipeline.
- **Automation rate:** Share of records processed without manual intervention.
- **Freshness:** How current data is (e.g. age since last update).
- **Pass rate:** Share of records that pass validation checks.
- **Schema conformance:** Data conforms to expected structure.
- **Lead time:** Time from commit to production (DORA).
- **Change failure rate:** Share of deployments that cause failure (DORA).
- **MTTR:** Mean time to repair.
- **Error rate delta:** Change in error rate after automation (+ worse, − better).

Metrics: Gen AI & LLM

- **Faithfulness:** Output grounded in retrieved context (RAGAS; <0.8 investigate, 0.9+ acceptable).
- **Answer relevancy:** Output addresses the query (RAGAS).
- **Context recall:** Retrieved chunks contain needed info (retrieval quality).
- **Context precision:** Retrieved chunks are relevant, not noise (retrieval quality).
- **Citation correctness:** Citations match sources.
- **RAG:** Retrieval-augmented generation.
- **RAGAS:** Framework for RAG evaluation (faithfulness, relevancy, context recall/precision).

- **LLM-as-judge:** Second LLM scores outputs when ground truth is expensive.
- **Ground truth:** Reference labels or human judgements used to evaluate model outputs.
- **Human-in-the-loop:** Human review for promotion and high-stakes validation.
- **MLflow GenAI:** Evaluation harness for GenAI; per-record and aggregate scores.
- **Component-aware:** Separate retrieval metrics from generation metrics to diagnose regressions.

Metrics: Traditional ML

- **Precision:** Share of positive predictions that are correct; few false alarms.
- **Recall:** Share of true positives correctly flagged.
- **F1:** Harmonic mean of precision and recall.
- **Time-to-detect:** Latency from event to detection (e.g. anomaly flagged).
- **Classification accuracy:** Share of predictions that match ground truth.
- **Inter-rater agreement:** Consistency between human raters or model vs human (e.g. GICS/ESG).
- **Break detection rate:** Share of true breaks flagged (reconciliation).
- **False break rate:** Share of non-breaks incorrectly flagged.
- **Time to resolve:** Time from break detection to resolution.
- **Mapping accuracy:** Correctness of client→FIGI/CUSIP mappings (DL+).
- **Time to first mapping:** Latency to produce first mapping.
- **Override rate:** Share of AI suggestions overridden by humans.
- **MAE:** Mean absolute error; average magnitude of errors.
- **RMSE:** Root mean squared error; penalises large errors more.
- **R^2:** Proportion of variance explained by the model.
- **Directional accuracy:** Correct up/down forecast (e.g. beat, meet, or miss estimates).

Metrics: Overarching

- **Error allowance:** Allowed failures before pausing (e.g. 1000 per 1M at 99.9% SLO).
- **SLO (Service Level Objective):** Internal target for reliability or performance; e.g. 99.9% of requests succeed, or p95 latency under 3s.
- **Cost per inference:** Cost per AI prediction (e.g. $ per LLM call).
- **Token throughput:** Tokens processed per second.
- **Freshness target:** Target for how current data must be (e.g. data under 5 minutes old).

- **Gate:** Checkpoint that must pass before proceeding.
- **Guardrail:** Threshold that triggers action (e.g. pause at 80% error allowance).

Other Key Concepts

- **Measure, gate, ship:** Define success before shipping; not "ship and hope".
- **Champion–challenger:** Compare new vs baseline; promote only when primary meets or exceeds baseline and guardrails hold.
- **Canary:** Gradual rollout of a new model to a subset of traffic before full deployment.
- **A/B roll out:** Split traffic between current (A) and new (B) model; compare metrics before promoting.
- **Rollback:** Reverting to the previous version when metrics breach guardrails.
- **Bloomberg Trust Frame:** AI as actionable, trustworthy insights; measure quality, risk, auditability.
- **Pydantic:** Python library for schema validation; enforces structure on LLM outputs and rejects malformed responses.
- **GX (Great Expectations):** Data validation and tests-as-code.
- **dbt:** Data build tool for transformation and testing.
- **Airflow:** Workflow orchestration.